

Solution of m Linear Equations in n Variables

Definitions and goals

We study systems of m linear equations in n variables with $m > 1$ and $n > 1$. Each equation has the form $a_{i1}x_1 + \cdots + a_{in}x_n = b_i$ (for $i = 1, \dots, m$), and a solution is any n -tuple $(x_1, \dots, x_n)$ that satisfies all equations.

General form (display)

A typical equation in the system appears as the display equation below:

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n = b_i, \quad i = 1, 2, \dots, m.$$

This single display emphasizes the linear structure shared by all rows.

Matrix notation

Writing the coefficient matrix $A = [a_{ij}]$, unknown vector x and right-hand side b , we compress the whole system into the matrix equation

$$Ax = b$$

which captures all m scalar equations at once.

Example system

Consider this concrete example with $m = 3, n = 3$:

$$\begin{aligned}2x_1 - x_2 + 3x_3 &= 5, \\x_1 + 4x_2 - x_3 &= -1, \\-x_1 + 2x_2 + 2x_3 &= 3\end{aligned}$$

Or, as a short inline example: the first equation is $2x_1 - x_2 + 3x_3 = 5$.

Cases and solution types

- If $m < n$ (underdetermined) there are typically infinitely many solutions (free parameters).
- If $m = n$ (square) there may be a unique solution, none, or infinitely many depending on A .
- If $m > n$ (overdetermined) the system can be inconsistent; consistency requires that all equations agree.

Summary

- **Problem:** Find all $x \in \mathbb{R}^n$ with $Ax = b$.
- **Methods:** Gaussian elimination / row reduction on the augmented matrix $[A \mid b]$.
- **Outcomes:** unique solution, infinite family (parametrized), or no solution.

Questions or follow-up: run a worked Gaussian-elimination example or include geometric interpretation?